

The Trezo Engine

Agents, gates, formulas, strategies — and what it cannot yet do

Trezo is an autonomous trading platform: a fleet of scheduled agents that scan markets, propose trades, argue with each other through a central risk gate, and manage whatever survives.

This document describes what actually runs — every rule read from the deployed source, every threshold quoted at its real value. The final section lists what is missing, modelled or provably inert, because a reference that only describes the intended behaviour is the kind of document that lets a silent failure survive for months.

Revision commit eb271bb · 23 August 2026

Scope the agents service — scanners, risk, execution, position management

Source read from the deployed tree, not from design notes

Prepared by Nova, for Mike

— Contents

Seven parts. The first three describe the machine; the next two describe what it knows; the last two describe what it trades and where it is weak.

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HOW TO READ A NUMBER IN THIS DOCUMENT

Every threshold quoted here was read from the source at commit `eb271bb` and is written at its deployed value, not its documented intent. Where the two disagree — and in several places they do — the disagreement is called out rather than smoothed over. Constants appear in `monospace`; anything marked **INERT** is code that exists, looks live, and cannot currently fire.

1 How the engine is put together

A signal is born in a scanner, judged once by a central gate, routed to a broker by a single executor, and then owned by a monitor until it closes. Nothing skips a step.

1.1 The loop

Every trade in Trezo travels the same four stages, and the separation is deliberate: the component that *finds* an idea is never the component that *approves* it, and neither one places the order.

SCANNERS (11 agents) propose signals	-->	RISK MANAGER (single gate) approve / veto size the trade	-->	TRADE EXECUTION (single router) route to broker or paper engine	-->	POSITION MONITOR (60-second tick) stops, targets, trims, reconcile
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Scanners propose. Pattern detection, STMS, ORB, Extended, Crypto, Forex, the Options Scanner and the Dividend LT agent each publish a signal onto the message bus. A signal carries a ticker, a direction, a strategy name, a confidence score, and a proposed stop and target. None of them can act.

The Risk Manager is the only gate. It is event-driven — interval 0, so it never ticks on a timer — and wakes only when a signal appears. It applies every rule in Part 3, rewrites the geometry when a target is unrealistic, sizes the position, and emits either an `approve` or a `veto`. There is no second opinion and no override path.

Trade Execution routes. Also event-driven, it listens for `approve` and nothing else. Stocks and options go to Alpaca; crypto goes to the internal modelled engine. This is the only component in the fleet that decides *where* an order goes.

The Position Monitor owns what exists. Every 60 seconds it walks every open position: enforcing stops and targets, ratcheting profit locks, applying time stops, stepping out of winners, and reconciling the local ledger against what Alpaca actually reports. It is the busiest agent and the one with the most order authority.

1.2 Books and pockets

Capital is divided twice. The outer division is the **book** — an independent account with its own equity, its own settings and its own open positions. The inner division is the **pocket** — a market-type budget inside a book.

The distinction matters because of a specific failure. Before 2026-08-20 the position cap was counted platform-wide, so three books holding roughly ten positions each read as fourteen of fourteen slots used, and **516 entries were killed by another book's activity**. Book scope now resolves before anything counts, and a signal whose book cannot be resolved is refused rather than defaulted to the primary.

POCKET BUDGETS BY POSTURE

POSTURE	CRYPTO	STOCKS	OPTIONS	INCOME	FOREX	AUTO-SELECTED WHEN
growth	32%	42%	10%	10%	6%	equity < \$25,000
balanced	18%	32%	20%	25%	5%	\$25,000 – \$100,000
income	9%	18%	20%	48%	5%	equity ≥ \$100,000
velocity	40%	22%	22%	8%	8%	never — opt-in only

A pocket is a hard dollar budget: `budget = equity × split`, and a trade whose notional would exceed what remains is skipped, not shrunk. Below \$25,000 the budgets soften — stretched by 1.75× and capped at 60% of equity per lane — because a 10% pocket of a small account cannot buy a single share of some names.

Position slots are divided the same way. A book's `max_open_positions` is shared out in proportion to pocket size, an unfunded pocket gets **zero** slots, and every funded pocket keeps at least one. A hot crypto lane cannot occupy the stock pocket's chairs.

1.3 The bus and the registry

Agents do not call each other. Every message goes onto a bus, which fans it out to every *other* enabled agent's `on_message` handler and persists it once. The registry holds one entry per agent — enabled flag, last tick, tick count, message count, last error, and whether it is an **actor** or an **observer**.

Scheduling is APScheduler with three settings that matter:

- `max_instances = 1` — a tick can never overlap itself.
- `coalesce = True` — a backlog of missed fires collapses into one run rather than stampeding.
- `misfire_grace_time = None` — a tick that came due while the loop was blocked runs late instead of being dropped.

Each tick also carries a hard timeout of `min(max(2 × interval, 120), 900)` seconds. A hung await records an error against the agent and lets the next fire proceed; it does not wedge the fleet.

1.4 Fail-open is a design rule, not an accident

The Wheel Advisor, the broker-tradable filter, the book gate, the outcome gate, the ops watchdog and the activity log all **allow** when they encounter an exception, missing data, or a disabled flag. The rule is stated explicitly in the advisor's own header: *a new advisory component must never be able to freeze a lane that was working before it existed*.

THE COST OF FAILING OPEN

A silently-broken guard is indistinguishable from a guard with nothing to say. This is exactly how lane rules 3 and 4 stayed inert for months: the ex-date guard was being handed `None` on every single call because it read an attribute the object never had, so it allowed everything, and allowing everything is what a healthy guard looks like from the outside. Both now read from real sources — but the design trade-off remains, and it is the reason Part 7 of this document exists.

1.5 Paper and live

Real money has a single chokepoint. `_LIVE_EXECUTOR_AVAILABLE` is a module-level constant, hardcoded `False`. `live_trading_enabled()` returns `live_requested() AND _LIVE_EXECUTOR_AVAILABLE`, so it is **always false in this build** — setting `TRADING_MODE=live` in the environment changes nothing real. Every execution path is required to test that function rather than the raw mode string, and an unrecognised mode falls back to `paper`.

Alpaca orders are placed against a paper brokerage account, so they are real orders through a real API against real market data, with simulated fills. Crypto is fully modelled — Kraken Spot has no paper API, so that lane runs on the internal engine with a 5 bps slippage and 26 bps commission model, permanently.

2 The agent roster

Twenty-nine registered agents. Five may place orders; the other twenty-four propose, advise or observe. The distinction is enforced at registration.

Agents are instantiated once at boot by a bootstrap routine that is idempotent, and each is registered with a role. **Five carry the ACTOR role** — Options Scanner, Trade Execution, Position Monitor, Kindrip and Dividend Manager. Everything else is an observer. Three agents have interval 0 and never tick on a timer at all; they run purely off bus messages.

AGENT	TICK	WHAT IT IS RESPONSIBLE FOR	AUTHORITY	OFF SWITCH
SCANNERS — they propose, they never act				
pattern_detection	180s	Scores candlestick patterns into trade confidence and emits signals.	PROPOSE	pattern_enabled
stms_scanner	180s	Small-cap momentum: \$1-\$20 names already running on abnormal volume.	PROPOSE	stms_enabled
orb_scanner	120s	Opening-range breakout on the first five one-minute bars of eight liquid names.	PROPOSE	stms_enabled
extended_scanner	1800s	Multi-day swings: EMA50 pullbacks, breakout holds, gap continuations, stair-steppers.	PROPOSE	extended_enabled
crypto_scanner	180s	24/7 scanner emitting SWING, HODL, DCA and SCALP signals across the majors.	PROPOSE	crypto_enabled
forex_scanner	180s	Scores 10 FX pairs off Kraken 4h bars. Dormant when no venue can convert the signal.	PROPOSE	forex_enabled
dividend_lt	1800s	Sizes the long-term dividend lane from each book's income pocket, proposes ladder entries.	PROPOSE	income pocket < \$500
THE SPINE — judge, route, manage				
risk_manager	event	The single gate. Approves or vetoes every signal, sizes it, enforces scope and kill switches.	APPROVE / VETO	—
trade_execution	event	Routes approved signals — stocks and options to Alpaca, crypto to the paper engine.	ORDERS	auto_trade_enabled
position_monitor	60s	Stops, targets, profit steps, day-trade management, broker reconciliation. The busiest agent.	ORDERS	no master switch
options_scanner	1800s	Runs the Dividend Wheel, settles expiries, harvests option winners, surfaces option ideas.	ORDERS	wheel_auto_execute

AGENT	TICK	WHAT IT IS RESPONSIBLE FOR	AUTHORITY	OFF SWITCH
broker_truth	900s	Asks Alpaca what options it actually holds and makes the ledger agree. Closes only unambiguous rows.	READ	—
dividend_manager	21600s	Credits modelled distributions, reinvests them, runs a once-daily income accumulation buy.	ORDERS	—
kindrip	21600s	Runs due contributions into children's index accounts and invests them. Modelled only.	MODELLED	—
CONTEXT — they change what the spine believes				
adaptive_scope	600s	Reads regime and breaking news, adjusts posture and strategy scope within guard rails.	MUTATES SCOPE	autonomy_mode
market_sentiment	1800s	Pulls watchlist news, scores sentiment, flags material events. Runs 08:00-16:00 ET.	READ	window only
market_horizon	900s	Cross-asset pulse: stocks, crypto, gold, dollar, bonds, income ETFs.	READ	—
cycle_awareness	21600s	Tags tickers with earnings and ex-dividend context — IV crush, dividend capture.	READ	—
research	1800s	Warns ahead of earnings (10-day horizon) and ex-dividend dates (14-day).	READ	—
relay_ingest	300s	Drains external briefings, validates each against a schema, files them into shared memory.	READ	—
REVIEW — they measure and report				
portfolio_architect	21600s	Daily structural review: lane edge, bet size via optimal-f, allocation via HRP, regime break via CUSUM. Proposes, never changes a rule.	READ	—
strategy_discovery	3600s	Win rate, profit factor and expectancy per book; flags a review every 25 trades.	READ	—
exit_advisor	300s	Tracks each position's peak unrealised P&L and alerts on a giveback of 30% or more.	READ	—
exit_advisor_options	300s	Option exit alerts: contract-count targets and a 39/30/25% drawback ladder.	READ	—
tax_optimizer	1800s	Tracks the tax impact of executed trades; nudges on retirement and employer match.	READ	—
OPERATIONS — they watch the machine, not the market				

AGENT	TICK	WHAT IT IS RESPONSIBLE FOR	AUTHORITY	OFF SWITCH
ops_watchdog	300s	Registry versus expected roster, per-agent silence, ops relay drain, broker route audit.	READ	—
book_health	300s	Per-book integrity: unmanaged notional, positions past their own stop, halts that outlived their cause.	READ	—
archivist	900s	Backs up activity logs and book state hourly to storage, weekly to Dropbox.	READ	TREZO_ARCHIVE_ENABLED
user_support	event	Registered stub. Both handlers return empty. INERT	NONE	—

SIX REGISTERED AGENTS ARE NOT SILENCE-MONITORED

The ops watchdog compares the registry against a hand-maintained list of expected agents, and that list holds 23 names against 29 registered. **forex_scanner**, **portfolio_architect**, **dividend_lt**, **broker_truth**, **book_health** and **archivist** can therefore go quiet without anything noticing. Given that two books once ran for six days with no watchlist and logged zero scanner messages daily — and nothing alerted — this gap is worth closing.

TWO COMMENTS IN THE SOURCE ASSERT THE OPPOSITE OF PRODUCTION

The position monitor states that the re-evaluator is "*master-flagged OFF, so this is a no-op*"; it is on, with 395 live lines behind it. The registry description for the forex scanner says "*disabled by default until data source is wired*"; it defaults on. Both are the kind of comment that makes a reader confident about the wrong thing.

2.1 Not an agent, but load-bearing

MODULE	WHAT IT IS	STATUS
activity_log	Fail-open JSONL ledger of every gate decision. Roughly 40 call sites.	Live. TREZO_ACTIVITY_LOG=0 disables.
reevaluator	395-line position re-evaluation engine, called by the position monitor.	Live behind TREZO_REEVAL_ENABLED.
base	The <code>Agent</code> abstract class and the message type.	Infrastructure.

3 The gates

Roughly forty distinct rules can stop a trade. They are grouped here by what they protect against, because that is the only useful way to reason about them.

A gate is not a preference. Most of these exist because something went wrong once and the number is the scar. Where that history is known it is stated, because a threshold without its reason is a threshold somebody will "optimise" later.

3.1 Capital and concentration

RULE	VALUE	WHAT IT PROTECTS AGAINST
A book will not open a new ticker once it holds its own maximum open positions	<code>max_open_positions</code> default 3	One book's count judging another's — the failure that killed 516 entries in a day
Each pocket gets only its proportional share of slots; unfunded pockets get zero, funded pockets keep at least one	<code>max(1, round(cap × alloc / total))</code>	A hot lane occupying another lane's chairs
Each market type has a hard dollar budget; a trade is skipped once it is spent	<code>equity × POSTURE_SPLIT</code>	Any single lane eating the whole account
Per-position notional cap, scaled by account size	<\$10k 15% · <\$25k 30% <\$100k 25% · ≥\$100k 15%	Low-volatility names with tiny stop distances taking ~58% of capital in one order
Total deployment ceiling across everything	<code>1.25 × equity</code>	Unbounded margin leverage
A single crypto coin cannot exceed a share of equity, summed across all its open rows	10%	Cross-day accumulation quietly concentrating the book in one coin
Crypto accumulation adds must be spaced apart in time	18 hours	"Buy the dip" firing every scanner tick instead of across days
The same ticker cannot be approved twice while still open	TTL 2 hours	One name approved three times in a morning
Lottery-ticket option structures capped as a share of options capital	3%	Allocation creep into low-probability plays
Below the hard-pocket floor, budgets stretch and then cap	<code>×1.75, max 60% of equity</code>	A pocket too small to buy one position at all

3.2 Liquidity and tradability

RULE	VALUE	WHAT IT PROTECTS AGAINST
Average daily volume floor, set per strategy	STMS 100k · ORB 500k Extended 500k · Wheel 500k global default 250k	Unfillable names. The old platform-wide 1M floor vetoed everything but the mega-caps
Bid/ask spread wider than the limit blocks the trade	0.5%	Slippage on illiquid names
No live quote <i>and</i> no recent trade reads as a halt	tape stale > 20 min	Trading into a halt — while tolerating a feed that is routinely empty
If the last bar is far from the quoted mid, the spread is unmeasured rather than wide	2% divergence	A "10.62% spread" artifact that produced 166 false vetoes in one day
Scalps are reserved for liquid capitalisation tiers	micro and unknown excluded	Scalping a name whose spread eats the entire edge
Small-cap momentum screen	\$1–\$20 · +10% move 5× relative volume · float < 20M	Wrong-shaped names entering the momentum lane
Dividend quality screen	payout ≤ 70% · yield ≥ 1.5% fund volume ≥ \$2M/day	Dividend traps and untradeable funds

THE PRICE FLOOR CANNOT FIRE

`MIN_PRICE = 0.0`, and every per-strategy override also sets it to zero. The check `if price < min_price` therefore never triggers — a stock at a hundredth of a cent passes. This is deliberate ("never gate on price"), but three separate comments still describe a live \$5 floor, including one in the risk manager. The behaviour is intended; the documentation around it is not.

3.3 Price extension and timing

RULE	VALUE	WHAT IT PROTECTS AGAINST
Will not chase a stock too far from its 20-day average, measured in ATR	$4.0 \times \text{ATR}(14)$	Buying the top of a parabolic move
Longs blocked when SPY <i>and</i> QQQ are both below session VWAP; shorts when both are above	cache 120s	Fighting the broad tape
Targets capped at what the name actually moves, then floored	$1.5 \times \text{ATR}\%$, floor 0.6%	"Waiting money" — a 10% target on a name that moves 2%
Intraday time stop, and stagnation stop	90 min hold · 75 min at <0.25R	Dead money occupying a slot. Crypto is deliberately exempt
End-of-day force exit for intraday strategies	19:45 UTC	Overnight gap risk on a day trade
Multi-day swing time stop	7 days	Swings becoming accidental investments

RULE	VALUE	WHAT IT PROTECTS AGAINST
Profit trail-to-lock once a winner has run	stocks +3% · scalps +2% giveback 30%	Round-tripping a winner back to breakeven
Opening-range quality band	10% – 80% of daily ATR	Breaking out of a range that is too tight, or already exhausted
Signals older than their deadline are auto-vetoed	300s in practice	Queue buildup when data APIs lag

THE STALENESS TIERING IS UNREACHABLE

The intent is that urgent signals expire in 60 seconds and ordinary ones in 300. The thresholds still test `tcs ≥ 700` and `tcs ≥ 500` — but the confidence score was rescaled to 0–100 in July 2026. Both bands are impossible, so every signal falls through to a flat 300 seconds. This is the same rescale-miss that was found and fixed elsewhere in the codebase; the fix was never applied here.

3.4 Risk per trade and portfolio limits

RULE	VALUE	WHAT IT PROTECTS AGAINST
Kill switch — daily realised loss	3% of day-start equity	Bleeding out in a single session
Kill switch — weekly realised loss	6% of week-start equity	A bad week compounding
Kill switch — consecutive losses, per book	3	Tilt, or a broken thesis repeating
Kill switch — broker rejects on a rolling window	3 in 60 min	A reject storm. The old session-scoped version once silenced a whole weekend
Kill switch — execution quality	3 fills worse than 75 bps	Trading into a market that will not give you your price
Risk per trade, as a fraction of equity	default 5%, clamped 0.05%–25%	Outsized single-trade loss
Reward-to-risk floor — the trade is rejected if the target is not worth the stop	1.5, clamped 0.3–3.0	Taking setups that need a 67% win rate to break even
Reward:risk harmoniser — whichever layer compressed the target, the stop follows	stop floor 0.4%	Every approval dying at "reward:risk 0.9 below 1.5" after an unrelated target cap
Crypto net-edge gate — the target must clear round-trip cost	+0.01% over 0.62%	Trading setups that are fee-negative before they start
Confidence floor, with additive bumps for hostile conditions	base 70 · earnings +5/+15 probation +10 · crowding +3/+6/+9	Taking marginal signals when the context is against you
Regime posture may tighten stops but never loosen them	multiplier 0.5 – 1.0	A market read quietly increasing risk

RULE	VALUE	WHAT IT PROTECTS AGAINST
Holdings cache is deliberately shorter than the monitor tick	45s vs 60s	A stale holdings set phantom-closing 17 real positions, which happened once

ONE GATE STILL HALTS THE PLATFORM ON ONE USER'S STATE

The per-user daily loss limit vetoes signals **for every book** when any single user is over their limit. The veto text says so plainly, but this is the last place where one account's state freezes everything — the same category error that was fixed for position counts and for consecutive losses.

3.5 Asset-class policy

Policy describes the *instrument*, never the trader's preference. An unregistered asset class fails closed to `UNKNOWN_POLICY`: still watched, still stopped on the client side, never stepped out of, never adopted.

CLASS	NATIVE BRACKETS	STOP TYPE	FRACTIONAL	SESSION GATED	ADOPTABLE
Stock	yes	stop	no	yes	yes
Crypto	no	stop_limit (gtc)	yes	no	yes
Option	no	—	no	yes	yes
Forex	no	—	yes	no	yes
Fund	no	—	yes	yes	yes
Unknown	no	—	no	yes	no

The stop-order type defaults to empty, which fails closed, because the failure mode is invisible: an unconfirmed venue rejects the stop and the position *looks* protected while being naked. A one-contract option cannot be split at all — stepping requires `qty ≥ min_slice + min_remainder`.

3.6 Options-specific preconditions

Before the Options Scanner may fire, all of the following must hold: the market is open; `wheel_auto_execute` is on (it defaults **off**); the account is not halted — and that check *fails safe*, returning halted on error; and Alpaca's options approval level is sufficient for the structure.

STRUCTURE	APPROVAL LEVEL	THROTTLES
Wheel legs (CSP, covered call)	≥ 1	4-hour cooldown per user/underlying/strategy after a block
Same-day and long options	≥ 2	3 open · 8/day · 3/symbol · min move 0.8% · PDT floor \$2,000
Spreads	≥ 3	\$150 risk · 1 open · \$20 minimum credit

Wheel collateral and duration are capped by posture: **growth** 25% of the pocket, 1 open CSP, 21 DTE; **balanced** 40%, 2, 35 DTE; **income** 50%, 3, 45 DTE; **velocity** 25%, 2, 10 DTE. Target DTE

deliberately picks one day under the cap so weeklies qualify.

4 The formulas

Every number the engine computes to decide a size, a price, a target or a probability. Written so they can be checked by hand.

4.1 Trade geometry

Geometry is the relationship between entry, stop and target. Trezo treats it as a first-class object because most bad trades are bad geometry, not bad selection.

```
reward_to_risk      RR = target_pct / stop_pct

break-even win rate  needed_win_rate = 100 / (1 + RR)
                    1:2    -> 33.3%      1:0.35 -> 74%
```

Verdict bands: $RR \geq 2.0$ healthy, ≥ 1.5 workable, ≥ 1.0 thin, below 1.0 upside-down. The floor for acceptance is 1.5.

A **rescale drift detector** watches for the case where a stop and a target are both adjusted but by different multipliers, silently degrading the trade. It reports when the ratio moves by more than 0.02 and never blocks.

The **harmoniser** runs last, after every other layer has had its say:

```
if target_pct / stop_pct < min_reward_risk:
    stop_pct = max( target_pct / max(min_reward_risk, 0.1), 0.004 )
```

4.2 Position sizing

```
stop_distance = |entry_price - stop_price|
eff_risk      = clamp(risk_pct, 0.0005, 0.25)
risk_usd      = equity * eff_risk
risk_qty      = risk_usd / stop_distance

notional_cap  = equity * position_pct_for_equity(equity)
if buying_power > 0: notional_cap = min(notional_cap, buying_power)

qty = risk_qty
if qty * entry_price > notional_cap: qty = notional_cap / entry_price
if asset is not crypto:             qty = floor(qty)      # reject if < 1 share

actual_risk   = qty * stop_distance
reward_risk   = |target_price - entry_price| / stop_distance # must clear 1.5
```

The sizer rejects outright on non-positive equity, non-positive prices, a zero stop distance, or a share count that rounds below one.

4.3 Volatility and ATR-derived geometry

```
true range  TR = max( high - low, |high - prev_close|, |low - prev_close| )
ATR         = mean(TR over the last 14 bars)

overextension stretch = (close - SMA20) / ATR14          veto if |stretch| > 4.0

scalp geometry atr_pct  = ATR14 / last_close
              stop_pct  = max(0.004, 0.8 * atr_pct)
              target_pct = max(0.006, 1.0 * atr_pct)

realistic cap cap = max(0.006, 1.5 * atr_pct)
              cap = min(cap, learned_achieved_move_pct)
              target_pct = min(target_pct, cap)
              stop_pct  = max(cap / min_reward_risk, 0.5 * atr_pct, 0.004)
```

The *learned achieved move* term is what stops the engine setting targets the name has historically never reached.

Realised volatility for option pricing uses **Yang-Zhang** by default, which accounts for overnight gaps as well as intraday range, falling back to close-to-close:

```
o_i = ln(open_i / close_{i-1})      c_i = ln(close_i / open_i)
k   = 0.34 / (1.34 + (m+1)/(m-1))
var = Var(o) + k * Var(c) + (1 - k) * RogersSatchell_var
sigma_annual = sqrt(var * 252)      clamped to [0.15, 2.00]
```

4.4 Capitalisation tiers

A stop that is right for a \$3 trillion company is wrong for a \$200 million one. Every strategy's geometry is scaled by tier as the last step:

TIER	MARKET CAP	STOP ×	TARGET ×	SCALP ALLOWED
mega	≥ \$200B	0.80	0.65	yes
large	≥ \$10B	0.90	0.75	yes
mid	≥ \$2B	1.00	0.80	yes
small	≥ \$300M	1.25	0.80	yes
micro	< \$300M	1.60	0.80	no
etf	—	0.85	0.70	yes
unknown	data miss	1.00	0.85	no

4.5 The confidence score

TCS — Trade Confidence Score — is computed in exactly one place, from five weighted components, and consumed everywhere else as a threshold.


```

technical = (pattern_score / 100) * 300
options   = 250                                if 30 ≤ iv_rank ≤ 60
           = 250 * (iv_rank / 30)                if iv_rank < 30
           = 250 * max(0, (100-iv_rank)/40) if iv_rank > 60
           = 100                                if iv_rank unknown
fundamental = 200 if catalyst_today else 80
rr          = clamp(rr_ratio * 60, 0, 150)        (110 when unknown)
market      = 50 + 30*(spy_trending_up) + 20*(confluence ≥ 30)

TCS = clamp( round( (technical + options + fundamental + rr + market) / 10 ), 0, 100 )

```

The structural reward:risk that feeds it is measured from real swing structure rather than a fixed percentage — risk is the distance to the recent swing low (floored at 0.75 ATR), reward is the distance to the swing high or two ATR, whichever is larger.

4.6 Expectancy and R-multiples

```

R          = pnl / |initial_risk|                initial_risk = |entry - stop| * qty
expectancy = mean(R)
dispersion = sample standard deviation of R
win_rate   = 100 * count(R > 0) / n
blown      = count(R < -1.0)                     # exit failures, not strategy outcomes
expectunity = expectancy * opportunities_per_period

profit_factor = sum(winning P&L) / |sum(losing P&L)|

```

The `blown` count is the useful one operationally: an R worse than -1 means the stop did not do its job, which is an execution problem rather than a strategy problem, and the two need different fixes.

4.7 Bet sizing — optimal f

Trezo uses Ralph Vince's optimal f rather than classic Kelly. The two answer the same question; f works directly from the trade list and the worst loss.

```

HPR_i = 1 + f * ( trade_i / |biggest_loss| )      # zero at f=1 on the worst trade
TWR    = product of all HPR_i                    # None if any HPR ≤ 0 (ruin)
G      = TWR ^ (1 / n_trades)                     # growth per trade

f*     = the f in {0.001 ... 1.0} that maximises G
f_dollar = |biggest_loss| / f*                    # dollars backing one unit

```

A companion routine inverts the question: *what is the largest f whose simulated path never exceeds a drawdown I can tolerate?*

WHAT OPTIMAL F CURRENTLY SAYS ABOUT THIS BOOK

Run over 196 real closed trades, $f^* = 0.001$ — the floor of the search. The geometric mean is below 1.0 at every fraction, meaning raising the bet size multiplies the loss rather than the return. The record is 98 wins and 98 losses for $-\$387$. That is a measurement of a partly-broken system rather than a verdict on the strategies, but no positive-edge measurement exists to set against it.

4.8 Allocation — hierarchical risk parity

HRP allocates across lanes without inverting a covariance matrix, in three stages.

```
1. distance      d(i,j)    = sqrt( 0.5 * (1 - corr(i,j)) )
                  dbar(i,j) = euclidean distance between columns i and j of d
                  then average-linkage agglomerative clustering on dbar

2. quasi-diagonal order = the leaf order implied by the merge tree

3. recursive bisection  w_i    = (1/var_i) / sum(1/var_j)      within a group
                        V_g    = w' * COV * w                group variance
                        alpha   = 1 - V(g0) / (V(g0) + V(g1))
                        scale g0 by alpha, g1 by (1 - alpha)
```

The estimate is only reported as trustworthy when there are at least $N(N+1)/2$ observations. HRP is return-blind — it will hand a near-idle lane most of the capital — so it is explicitly paired with the expectancy screen rather than used alone.

4.9 Statistical significance

These exist so that "this strategy works" can be challenged with a number.

```
Sharpe              SR = mean(returns) / stdev(returns)

Probabilistic Sharpe PSR = Phi( (SR - SR_benchmark) * sqrt(n-1)
                               / sqrt(1 - skew*SR + ((kurt-1)/4) * SR^2) )

Expected max Sharpe SR* = sd(trial_sharpes) *
                        [ (1-g) * Phi^-1(1 - 1/N) + g * Phi^-1(1 - 1/(N*e)) ]
                        g = 0.5772 (Euler-Mascheroni)

Deflated Sharpe     DSR = PSR(returns, benchmark = SR*)
```

The deflated Sharpe is the honest one: it discounts a good result by how many strategies were tried before it appeared. There is also an **implied precision** solver — the minimum win rate a given stop/target geometry needs to hit a target Sharpe — and a bootstrap that reports whether the mean outcome's 95% interval excludes zero.

4.10 Structural break detection

A CUSUM test on log prices, used by the portfolio review to ask whether the regime a strategy was calibrated in still exists.

```
y_t      = ln(price_t)
sigma_t  = sqrt( sum (y_i - y_{i-1})^2 / (t-1) )
S(n,t)   = (y_t - y_n) / ( sigma_t * sqrt(t - n) )
c(n,t)   = sqrt( 4.6 + ln(t - n) )                # 5% one-sided critical value

break detected if max over n of [ |S(n,t)| - c(n,t) ] > 0
```

Measured false-positive rate on random walks is 3.5% against a nominal 5%, so it errs conservative.

4.11 Trading costs

```
half_spread = ((ask - bid)/2) / mid
round_trip  = 2*(fee_bps/10000) + 2*half_spread + 2*(slippage_bps/10000)

viable      = target_pct ≥ round_trip * safety_multiple      # default 2.0
```

Model constants: 5 bps slippage per side, \$0 stock commission, 26 bps crypto commission. Crypto round trip is therefore 0.62%, which is why the crypto lane carries its own net-edge gate. A zero cost estimate is treated as a missing measurement, not a free trade.

4.12 Option pricing

Black-Scholes, European, no dividend term, no rho, risk-free rate hardcoded at 4.3%.

```
d1 = ( ln(S/K) + (r + sigma^2/2) * T ) / ( sigma * sqrt(T) )
d2 = d1 - sigma * sqrt(T)

call = S*N(d1) - K*e^(-rT)*N(d2)      delta_call = N(d1)
put  = K*e^(-rT)*N(-d2) - S*N(-d1)   delta_put  = N(d1) - 1

gamma = phi(d1) / ( S * sigma * sqrt(T) )
vega  = S * phi(d1) * sqrt(T) / 100   # per 1 IV point
theta = annual_theta / 365            # per calendar day
```

Implied volatility, where a live premium is available, is solved by bisection on [0.01, 5.0] to a tolerance of 1e-5.

The variance premium — the whole reason to sell options — is measured as:

```
premium_ratio = IV / RV
verdict: RICH  if ratio ≥ 1.20      FAIR if > 1.00      CHEAP otherwise
```

THE WHEEL'S VOLATILITY EDGE IS CURRENTLY ZERO BY CONSTRUCTION

Because implied volatility is estimated *from* realised volatility, the modelled premium ratio is identically 1.0. The wheel cannot presently distinguish an expensive option from a cheap one, and whatever it earns comes from direction and assignment rather than from selling volatility — which is nominally its entire business. Live quotes are fetched, but only *after* the go/no-go decision has been made on the modelled premium.

5 The strategies

Nine lanes. Each one states its trigger, its exit, the pocket it spends from, and the conditions under which it is not allowed to trade at all.

Above the individual strategies sit three layers that shape all of them: a **regime classifier** that reads SPY into one of six states, an **adaptive posture** layer that converts a regime into a stop multiplier and a confidence bump — and can only ever tighten, never loosen — and a **selector** that runs a per-stock bake-off between eligible strategies, dropping any with a negative track record on that specific ticker.

5.1 STMS — small-cap momentum

Hunts small-cap stocks already running hard on abnormal volume during the session, and rides the continuation. The watchlist is dynamic, built from the day's top gainers filtered to the price band, with a static seed as fallback.

Entry	Price \$1.00-\$20.00, daily move $\geq +10\%$, relative volume $\geq 5\times$ the 20-day average. Then separately: float under 20M shares, a news catalyst, and a structural chart read — a pole of at least $+12\%$, a pullback between -2% and $+15\%$, and a recent low above the pole's midpoint.
Exit	Stop 5% below entry. Target $+10\%$ on half; trailing stop on the remainder. Subject to the intraday force exit.
Pocket	Stocks.
Guards	Weekdays, 11:00-21:00 UTC (a deliberately wide union so daylight saving never matters). Liquidity floor lowered to 100k for this lane alone. Confidence floor 75.

5.2 ORB — opening range breakout

Takes the high and low of the first five one-minute bars as the opening range and trades a confirmed break of it, on a fixed list of eight liquid names.

Entry	Two consecutive one-minute closes on the same side of the range. Range height must be 10%-80% of the daily ATR. Breakout-bar volume is scored, not required.
Exit	Stop at the far side of the opening range; target at 2R.
Pocket	Stocks.
Guards	Windowed. Note: the docstring and the constants disagree — the prose describes 9:35-10:30 ET, the deployed constants run 8:30 AM ET (pre-market) as "best", 10:30 as "reduced", closing at noon. The constants are what runs.

5.3 Extended — multi-day swing

The only lane that holds across sessions. Four continuation setups on daily bars across a twelve-name mid-cap watchlist; the highest-scoring one wins.

SETUP	TRIGGER	STOP / TARGET
EMA50 pullback	Price above a rising 50-EMA, a bar low within 3% of it in the last six, last bar closes up and back above	5–10% / 1.8× stop
Breakout hold	Close above the 33-bar high excluding the last three, still no more than 8% above it	5–9% / 1.8× stop
Gap continuation	A gap up of 4% or more in the last five bars, still unfilled	7% / 10%
Stair-stepper	Four consecutive five-bar blocks of both rising highs and rising lows, total run 8–60%	7% / 10%

Profit ladder ratchets at +4% (lock breakeven), +7% (lock +3%), +10% (lock +6%). Time stop at 7 days. **Exempt from the intraday force exit** — it is the one lane designed to hold overnight. No new entries before 18:00 UTC on an FOMC decision day.

5.4 Crypto — four modes, one scanner

Long only, 24/7, four mutually exclusive modes evaluated in priority order. Every entry must clear the fee-edge gate before anything else is considered.

MODE	ENTRY	STOP / TARGET	CHARACTER
SWING	Bollinger width > 2.0%, RSI 48–72, volume ratio ≥ 0.8×	5% / 12%	Overrides per-coin geometry entirely
HODL	RSI < 25	–35% / never	Trail-to-lock engages at +40%, trailing 20% below the high
DCA	RSI < 40 <i>and</i> bouncing — RSI up on the bar and close above the prior close	per-coin	The bounce requirement is what stopped it catching knives
SCALP	Bollinger width < 25%, RSI 40–68, volume ratio ≥ 0.4×	0.6× / 0.5× per-coin	See the note below

Profit ladders lock a floor as the trade runs: SWING ratchets at +0.8%, +1.0%, +1.8%, +3.0%, +5.0%, +10%; DCA uses the same rungs minus the last. Only HODL and DCA may add to the same ticker across days.

SCALP'S GEOMETRY IS QUIETLY DISTORTED

The stop is multiplied by 0.6 and the target by 0.5, which turns a designed 1:2 into 1:1.67. It went unnoticed for weeks — the only trace was four stop-outs clustering at –1.9%. The geometry guard now *reports* the distortion but deliberately does not block it.

5.5 The Dividend Wheel

Sell cash-secured puts on quality dividend names; if assigned, sell covered calls above cost basis; repeat. Premiums are modelled unless a live quote refines them.

CSP entry	Strike at $\text{spot} \times 0.95$, at a target 30 DTE. Requires at least 22 candles and a positive credit. Decay override: if trailing monthly drift is -2% or worse, the strike is projected <i>down</i> to where the decay lands over the contract's life, with a 25% safety margin — never up.
CC entry	Strike at $\max(\text{spot} \times 1.05, \text{cost_basis} \times 1.01)$. If an ex-dividend date falls inside the contract's life the cushion widens to 7.5%, so the shares are not called away before the dividend is captured.
Recovery CC	For lots 15% or more underwater, a call may sit below cost basis, but only if all three hold: the strike is above <i>spot</i> ; premium per month exceeds the lot's trailing monthly bleed; and the credit is at least $\max(\$10, 0.2\% \text{ of lot value})$. No symbol is ever banned.
Pocket	Income.
Universe	A 22-name curated seed rotated daily by ordinal so one cheap name cannot win every cycle, plus dividend-named user watchlists, plus open option positions, plus a 60-name market-wide pool, plus anything the dividend screen passes.

THE WHEEL ADVISOR

An advisory gate at the end of the wheel path, added deliberately as a variable rather than a rewrite. It can only **subtract** — defer a leg or shrink it, never add or enlarge — and it **fails open** on any exception, missing data, or disabled flag. Four checks: a 5–45 DTE window, a ± 2 -day earnings blackout, the ex-date guard, and ledger collateral honesty. GROWTH-tier names can never wear a call.

5.6 Options structures

Eight defined-risk builders, all priced at 35 DTE with strikes as fixed offsets from spot, all carrying net position Greeks. These are *ideas surfaced* by the Options Scanner; the directional selector explicitly excludes them.

STRUCTURE	STRIKES	BIAS	MAX LOSS / MAX GAIN
Long call	1.02	bullish	debit / unbounded
Long put	0.98	bearish	debit / $\text{strike} \times 100 - \text{debit}$
Bull call spread	1.01 / 1.08	bullish	net debit / width – debit
Cash-secured put	0.95	income	collateral – credit / credit
Bull put spread	0.95 / 0.90	income	width – credit / credit
Bear call spread	1.05 / 1.10	income	width – credit / credit
Butterfly	0.97 / 1.00 / 1.03	neutral	debit / $(\text{mid} - \text{low}) \times 100 - \text{debit}$
Iron condor	0.95/0.90 + 1.05/1.10	income	wing – credit / credit

5.7 Cycle-aware strategies

Two strategies exist only as scoring biases inside the selector, with no dedicated module. **IV crush short** enters the pool when the volatility environment is high or earnings are one to seven days out. **Dividend capture long** enters when an ex-dividend date is two days behind to five days ahead.

Both deliberately **bypass the earnings-day confidence bump** — the bump exists to keep ordinary strategies out of an earnings window, and these two exist precisely to trade it.

5.8 Forex — declared, structurally dormant

Ten Kraken pairs on 4-hour candles, long and short, stop at 1.0× and target at 1.2× the 4-hour ATR. Every signal is vetoed at the risk gate because the broker has no FX venue and the position could only ever be modelled. The scanner detects this and stops emitting rather than manufacturing guaranteed vetoes — but it still ticks every 180 seconds across ten pairs. Either give it a venue or turn it off; running it as-is is overhead plus veto noise that hides real rejections.

5.9 The strategy library

Fifteen canonical strategy cards and a regime playbook (favour / reduce / pause / probation by regime) exist as a read-only knowledge base. **Only six map to live modules.** The other nine — post-earnings drift, pairs, sector rotation, VWAP reversion, relative-strength momentum, news catalyst, and others — are described and never traded. This is stated as a philosophy rather than a backlog: a handful of strategies understood deeply beats hundreds that cannot be supervised.

6 The Dividends (Long-Term) lane

The newest lane and the most rule-dense. A three-sleeve income structure whose behaviour is a pure function of the capital it is given — size buys mechanics, not edge.

6.1 Sizing

The lane takes a capital figure from its book's income pocket and splits it three ways. Everything downstream follows from that split.

```
L = capital * w_ladder      # the dividend ladder
W = capital * w_wheel       # the wheel sleeve
B = capital * w_buffer      # cash buffer

ladder_names = clamp( floor(L / 1000), 1, 15 )
csp_blocks   = floor( W / block_cost )          # block_cost default $3,500

income_monthly = L * 0.053/12 + csp_blocks * block_cost * prem_rate + B * 0.0375/12
expected_TR    = w_ladder*(0.053 + 0.045) + w_wheel*0.080 + w_buffer*0.0375
```

The guard rails are **hard**: they raise an error rather than silently clamping, because a silently clamped input is an input the owner thinks was accepted.

INPUT	PERMITTED RANGE	WHY THE BOUND EXISTS
Ladder weight	50% – 90%	Below half, it is not an income ladder
Wheel weight	0% – 40%	The wheel's added return is unproven; it cannot dominate
Buffer weight	3% – 20%	Assignment needs cash that is not already spoken for
Wheel delta	0.15 – 0.40	Above 0.40 the wheel is a directional bet wearing an income costume
Target return	5% – 25%	Outside this band the target is not a plan
Minimum capital	\$500	Below it, no sleeve can be constructed

6.2 What capital unlocks

UNLOCK	CONDITION	WHAT IT CHANGES
U1	Wheel sleeve covers one block	The wheel becomes available at all
U2	Two or more CSP blocks	Premium can be laddered across expiries rather than concentrated
U3	Twelve or more ladder names	Per-name cap tightens from 20% to 10% — enforced, not advised
U4	\$50,000 capital	Selectivity: the lane can afford to decline marginal names

6.3 The per-name state machine

Every name in the lane sits in exactly one state, and the state decides what the engine is permitted to do with it. The precedence is deliberate: cash-reserved is checked first, assignment provenance second, so a name can never be in two states at once.

STATE	CONDITION	WHAT IS PERMITTED
FRACTIONAL	Fewer than 100 shares	Dividends only. No options of any kind.
LOT_READY	100+ shares, HIGH_YIELD tier	Dividends, and covered calls are eligible.
LOT_HELD	100+ shares, GROWTH tier	Dividends only. Never a covered call.
CASH_SECURED	Cash reserved for a put, no shares	Premium. No dividend — the shares are not owned.
ASSIGNED	Put exercised, 100+ shares	Dividends, and covered calls are eligible.

Transitions: a position that organically reaches 100 shares moves **FRACTIONAL** → **LOT_READY**, or to **LOT_HELD** if the name is a compounder. Exercise moves **CASH_SECURED** → **ASSIGNED**. An unknown or unverified tier is *not* call-eligible — silence is not consent.

6.4 The five lane rules

#	RULE	WHAT GOES WRONG WITHOUT IT
1	Two-state names. A wheel name is either cash securing a put <i>or</i> shares wearing a call — sequential, never simultaneous. A CSP is refused on a name already holding a round lot.	The same capital counted twice, once as collateral and once as shares
2	Graduation at 100 shares. Round lots gate covered calls; fractional positions are dividend-only.	Writing a call against shares that do not exist
3	Ex-date guard. Never carry an in-the-money short call into an ex-dividend date. Permitted if expiration clears the date, or an out-of-the-money buffer above 1% exists. Where time value and dividend amount are known it applies the real test: exercise is irrational while time value exceeds the dividend.	Early exercise takes the dividend the lane exists to collect
4	No covered calls on GROWTH names. Writing calls on a compounder sells away 4–6% a year of payout growth for a one-month premium.	Capping the exact upside the ladder is built to compound
5	Hard collateral reservation. CSP collateral is reserved cash and cannot double-count as ladder capital. Already-reserved collateral is subtracted before the question is asked.	An assignment the lane cannot fund

RULES 3 AND 4 HAD NEVER FIRED ONCE

Both read their inputs from an attribute the object did not carry, so both received `None` on every single call and allowed everything. A guard that allows everything looks exactly like a guard with nothing to say. They now read from real sources — but this is the clearest illustration in the codebase of what fail-open costs.

6.5 The entry screen

The screen decides what may enter the ladder at all. It runs different tests for operating companies and for funds, because the same question means different things.

FOR A COMPANY

```
yield          ≥ 1.5%
payout ratio ≤ 70% of earnings      (marked n/a for REITs, BDCs and funds)
dividend trend: an unbroken raise streak across everything visible,
                  up to a 10-year target, and no cut inside the record
```

A cut no longer disqualifies permanently. It disqualifies until it is **repaired**, which requires all three of:

```
recovered = latest annual dividend ≥ the pre-cut peak
rising    = raised every complete year since the trough
healed    = at least 3 complete years since the trough
```

Across a 120-name universe this admits six names that a flat no-cut rule excluded, and still excludes fifteen — every one of which is *still* paying less than it did before its cut.

FOR A FUND

A fund's distribution varies by design, so a raise streak is the wrong instrument entirely. The question is whether the distribution is funded:

```
fail if trailing distribution yield > trailing total return    # eating NAV
fail if any reverse split within 24 months
size floor: $2M average daily dollar volume                  # standing in for AUM
```

WHY THE FUND RULE MATTERS

Run the company rule over funds and it fails seven of eight covered-call ETFs, because a variable distribution is what those funds *are*. One fund paid 5.05, then 19.53, then 12.14 — option premium tracking volatility, read by the screen as a dividend cut. The funded test discriminates properly: measured live, JEPI paid 7.9% and earned 9.4% (pass), while NVDY paid 57.2% and earned 21.8% (fail).

TIERING AND CONCENTRATION

A passing name is **HIGH_YIELD** at or above 4.0% yield and **GROWTH** below it. The tier is what lane rule 4 hinges on. At selection time no more than two names per sector are admitted; REITs and BDCs collapse into one bucket because they are one factor; funds get their own bucket, so a failed sector lookup on an unrelated stock cannot evict a fund from the ladder.

6.6 The target-return dial explains, it does not actuate

Moving the target return changes no order, no strike and no allocation. It prints the arithmetic instead:

```
gap                = target_return - expected_TR
appreciation_required = ladder_growth + gap / w_ladder
premium_required    = base_premium_rate + gap / w_wheel / 12
implied_delta       = the delta that would produce premium_required

unreachable if implied_delta > 0.40          # the hard cap in the guard rails
```

The design note is blunt about why: a dial that chases a number would be the same mistake with a nicer interface. The readout names which guard rail blocks the path rather than quietly loosening it.

6.7 Income mode

```
draw = max( 0, min( actual_distributions_received, 0.90 * trailing_12mo_total_return ) )
```

By construction this never draws principal: the smaller of what was actually received and 90% of what was actually earned. Shortfalls come from the buffer.

7 Limitations

What the engine cannot do, cannot see, or does not really do despite appearing to. This section is the reason the rest of the document can be trusted.

Four categories, and they need different responses. A **data gap** is closed by buying or building a feed. A **modelling approximation** is closed by measurement. **Inert code** is closed by deleting it or wiring it up. A **deliberate deferral** is not a defect at all — it is a decision, and it is listed so that nobody re-discovers it as a surprise.

7.1 Data-source gaps

WHAT IS MISSING	WHAT DEPENDS ON IT
No option Greeks or implied volatility from any feed. Snapshots return quotes and bars only.	Every wheel strike, all eight option structures, the advisor's delta reasoning, the lane's delta table. Delta is modelled, never observed.
No historical option chains.	The wheel sleeve could not be included in the ladder replay at all — modelling it would have invented the number the exercise was meant to measure.
No point-in-time fundamentals. Nobody on this stack offers them.	Every backtest that screens on fundamentals carries lookahead bias. The three-month ladder replay is an anecdote with a number attached, not evidence.
The dividend payment series is not on the current data tier.	Now sourced from the broker's corporate-actions feed instead. This works, and it uncovered three separate feed defects — see 7.2.
Corporate-actions history begins in 2016. Ten complete years yield at most nine year-over-year comparisons.	A literal ten-year streak requirement is not yet expressible; the rule reads "unbroken across everything visible" and tightens on its own as history accumulates.
Fund fundamentals are not covered. An ETF returns 19 price technicals and not one company figure.	AUM is unmeasurable, so a dollar-volume liquidity floor stands in — named as a substitute in the code rather than dressed up as AUM.
No real macro source. Volatility-index and treasury reads fall through to an ETF proxy.	Macro context in scoring is approximate.
Top-of-book artifacts from the market-data feed.	Roughly 700 vetoes a day are data quality rather than signal quality: 199 "no live quote — possibly halted" on names that were not halted, 166 "spread too wide" from stale-book divergence, 327 "no price data".
No real fills, so no true slippage measurement.	Slippage is a 5 bps-per-side model. This cannot be closed with code — only with live execution.

7.2 Modelling approximations

THE APPROXIMATION	WHAT RESTS ON IT
Every option premium is Black-Scholes. European exercise, no dividend term, risk-free rate hardcoded at 4.3%, implied volatility proxied by realised volatility.	Every wheel go/no-go and every Greek. A live quote refines the strike and premium <i>after</i> the decision has been made.
The wheel's volatility edge is zero by construction. Because IV is derived from RV, the modelled variance premium is identically 1.0.	The wheel cannot tell an expensive option from a cheap one. Whatever it earns comes from direction and assignment, not from selling volatility.
Every calibration constant in the dividend lane is a placeholder for a measurement. The wheel's 8.0% total-return assumption is labelled unproven in the source; the entire delta table is unproven pending live chain data.	Expected return, monthly income, the target readout, and every projection the lane produces.
The FOMC blackout list is hand-maintained.	The Extended lane's only event gate. A stale list silently means no gate at all.
All trading windows are UTC approximations, deliberately widened to cover both daylight-saving states.	Half the year each window is an hour wider than intended. An accepted trade-off, stated as such.
Dividend distributions are modelled. No payment-frequency field exists, so everything defaults to quarterly.	This understates genuinely weekly and monthly funds. The live ex-date fetch is written but untested against production payloads.
Crypto fills are modelled permanently. The venue offers no paper API.	All crypto profit and loss to date — which is the only lane currently making money.
The crypto ladder replay knows entry, peak and exit — not the path between.	It cannot see a stop tripped early by a wiggle, so the ladder rungs it produced are optimistic by an unknown amount.
The equity profit ladder was deliberately not retuned on that replay, because crypto pays 0.62% round trip and equities are commission-free.	Extended's profit locks rest on untested rungs — honest, but untested. Borrowing a conclusion across cost models would have been worse.
The covariance estimate behind HRP cannot honestly be computed yet.	Portfolio-level allocation guidance. Reported with a trustworthiness flag rather than presented as fact.

7.3 Known-inert code

Code that exists, reads as live, and cannot currently fire. Each of these was found by looking, not by an alert — which is itself the finding.

WHAT IS INERT	CONSEQUENCE
Lane rules 3 and 4 had never fired. Both read an attribute the object did not carry.	The two rules protecting dividends from early exercise, and compounders from being called away, were silent. Now fixed.
The scheduler's intended start path has never executed. The caller passes two arguments to a function that takes none; every boot raises, is caught, and a fallback starts the engine.	One boot log signal has been lying since it was written.
The staleness tiering is unreachable — thresholds still test the pre-July confidence scale.	Urgent signals do not expire faster than ordinary ones. Every signal gets a flat 300 seconds.
The price floor cannot fire — every override sets it to zero.	Intended behaviour, but three comments still describe a live floor.
ORB and Extended still emit confidence on the old 0-1000 scale.	Their minimum-confidence floor can never bind, and their signals clear any 0-100 gate unconditionally. This is the same rescale-miss family as the staleness bug, still live in two modules.
Five modules are dead — roughly 734 lines. No test defends any of them; nothing has been deleted.	Cognitive load. One of them documents that the risk gate "should call it on every veto" — it never has.
Forty-three public functions are defined and never called, including a 63-line drift detector inside a very live file.	Deliberately left as groundwork rather than abandonment, but indistinguishable from abandonment without this note.
The forex lane burns cycles it cannot convert. It ticks every 180 seconds and every signal is vetoed — 465 vetoes in one day at peak.	Scheduler capacity, and veto noise that hides real rejections.
The market-wide wheel gate was a whitelist in a market-wide costume. The pool was market-wide; the gate asked a 40-name dictionary and fell back to a source rate-limited to five calls per build.	Any name not pre-listed effectively could not qualify, so the pool collapsed back to the curated list every tick. Fixed 2026-08-22.

WHAT IS INERT	CONSEQUENCE
Two books ran with no watchlist and logged zero scanner and zero risk messages, every day, for six days. Their last scanner-driven stock entry was the same minute on both books.	Two-thirds of the platform's stock lane was silently idle. Nothing alerted. Synchronised stops are configuration, not markets.

7.4 Deliberate deferrals

DEFERRED	STATED REASON
ORB credit spreads (section 7 of that design)	Needed structures the platform lacked at the time. The builders now exist, so this may be stale.
Intraday penny patterns — supernova, short squeeze, precise bull flag	All need an intraday feed. A structural daily-candle read substitutes.
Acting on the variance-premium measurement	The go/no-go is already made on the modelled premium. Acting on the live measurement is a behaviour change and belongs in a proposal, not a patch.
Nine of fifteen library strategies	A handful understood deeply beats hundreds that cannot be supervised.
Live brokerage execution	Scaffolded and inert behind a go-live checklist. A planned phase, not an oversight.
Crypto short side, and real venue order placement	Real money — requires an explicit decision, not an inference.
Broker-routed partial closes on bracket-attached positions	Requires a cancel → partial sell → re-submit pattern that has bitten once already.
Trusting the dividend-lane ranking	It adds 0.70 percentage points on a three-month sample across fourteen names — indistinguishable from noise. The screen is doing the work; the ordering is not, at least not yet.
The twelve-month verdict gate	Ninety days proves plumbing. Twelve months is the minimum for a total-return verdict, and no dial moves that gate.

7.5 Two things the record says that no single note frames as a limitation

ONE LANE IS CARRYING THE PLATFORM, AND ITS FILLS ARE MODELLED

Lifetime realised: stocks **+\$53 across 318 trades**; crypto **+\$11,624**; options **\$0** on five positions that have never been closed; forex **+\$4** across 19 trades on a venue that no longer exists. The lane producing essentially all of the profit is the one whose fills are simulated by the platform's own engine rather than by a broker. That does not make the result wrong. It does mean it is the least externally validated number in the system, and it should not be treated as the most reliable one.

THE WHEEL HAS NO REALISED TRACK RECORD

The decision chain works — signals are generated, gates are passed, orders are submitted. They then fail at the last inch: one account lacks options approval, which is a dashboard toggle; two others die on "no listed put contract near target". Five real option positions exist, all adopted rather than opened, at credits between \$0.02 and \$0.19, none of them closed. Every statement about the wheel's expected return in this document is a model output, not a measurement.

7.6 How to read this section in six months

The pattern worth noticing across 7.3 is that **none of those failures produced an error**. The gate that admitted nothing looked like a strict gate working. The rule that never fired looked like a rule with nothing to say. The lane with no watchlist looked like a quiet market. In every case the system was reporting success while doing nothing.

The design rule that everything fails open is correct — a new advisory component must never be able to freeze a lane that worked before it existed. But its cost is exactly this class of silence, and the only defence is to periodically ask each guard when it last actually fired. A guard that has never fired in production is either unnecessary or broken, and it is worth knowing which.

A Constants quick reference

Every number that changes behaviour, in one place, at its deployed value.

A.1 Risk and capital

CONSTANT	VALUE	EFFECT
Risk per trade	5% (clamp 0.05% – 25%)	Dollars risked per position
Reward:risk floor	1.5 (clamp 0.3 – 3.0)	Below this the trade is rejected
Absolute stop floor	0.4%	The harmoniser will not tighten past this
Max open positions per book	3	New-ticker cap
Total deployment ceiling	1.25 × equity	Leverage bound
Per-position notional cap	15 / 30 / 25 / 15%	By equity band, smallest to largest
Crypto per-coin cap	10% of equity	Summed across all open rows
Crypto accumulation cooldown	18 hours	Spacing between adds
Approval TTL	2 hours	Duplicate-approval guard
Hopeful-bucket cap	3% of options capital	Low-probability structures
Small-account pocket stretch	1.75×, cap 60%	Below \$25,000 equity

A.2 Kill switches

TRIGGER	THRESHOLD
Daily realised drawdown	3% of day-start equity
Weekly realised drawdown	6% of week-start equity
Consecutive losses, per book	3
Broker rejects	3 within 60 minutes
Slippage breaches	3 fills worse than 75 bps
Per-coin daily loss bench	10% of coin budget (advisory)

A.3 Filters and windows

FILTER	VALUE
Max ATR stretch from SMA20	4.0
Max bid/ask spread	0.5%
Global average-volume floor	250,000

FILTER	VALUE
Per-strategy volume floors	STMS 100k · ORB / Extended / Wheel 500k
Tape freshness	20 minutes
Stale-book divergence	2%
Intraday max hold	90 minutes
Stagnation stop	75 minutes below 0.25R
End-of-day force exit	19:45 UTC
Swing max hold	7 days
Signal staleness	300 seconds (effective)
Confidence floor	70 base · 35 crypto · 75 STMS
Max confidence bump	15
Stop multiplier bounds	0.5 – 1.0 (tighten only)
Book-scope cache TTL	45 seconds

A.4 Options and the wheel

CONSTANT	VALUE
Target DTE	30
CSP / CC out-of-the-money offset	5% (7.5% across an ex-date)
Advisor DTE window	5 – 45
Earnings blackout	±2 days
Posture collateral caps	growth 25% · balanced 40% · income 50% · velocity 25%
Posture open-CSP caps	1 / 2 / 3 / 2
Posture DTE caps	21 / 35 / 45 / 10
Decay override trigger	–2% monthly drift, 25% safety margin
Recovery distress threshold	15% below basis
Recovery minimum credit	max(\$10, 0.2% of lot value)
Minimum DTE for any option	7
Risk-free rate	4.3%
Implied-volatility clamp	15% – 200%
Variance-premium bands	rich ≥ 1.20 · fair > 1.00 · cheap otherwise

A.5 The dividend lane

CONSTANT	VALUE
Ladder / wheel / buffer bounds	50–90% / 0–40% / 3–20%
Wheel delta cap	0.15 – 0.40
Target return bounds	5% – 25%
Minimum lane capital	\$500
Max ladder names	15
Capital per ladder name	\$1,000
CSP block cost	\$3,500
Per-name cap	20%, tightening to 10% at U3
Assumed ladder yield / growth	5.3% / 4.5%
Assumed wheel total return	8.0% – unproven
Assumed buffer yield	3.75%
Income-mode draw ceiling	90% of trailing total return

A.6 The dividend screen

CONSTANT	VALUE
Minimum qualifying yield	1.5%
Growth / high-yield tier split	4.0%
Maximum payout ratio	70%
Raise-streak target	10 years (window-limited)
Cut lookback	10 years
Cut detection band	5% year-over-year decline
Cut heal period	3 complete years since the trough
Fund liquidity floor	\$2M average daily dollar volume
Fund AUM floor (unmeasurable on this tier)	\$100M
Reverse-split lookback	24 months
Special-dividend detection	3× the year's median payment
Fragment detection	below one third of the year's median
Payment-count tolerance band	±25% of the modal count
Implausible-growth guard	25% per year
Sector concentration cap	2 names

CONSTANT	VALUE
Screen cache	7 days · 25 lookups per build

A.7 Cost model

CONSTANT	VALUE
Slippage, per side	5 bps
Stock commission	\$0
Crypto commission	26 bps
Crypto round trip	0.62%
Crypto minimum net edge	+0.01% over round trip
Viability safety multiple	2.0 × round-trip cost

A CLOSING NOTE ON NUMBERS

Most of the thresholds above are scars. The 45-second cache is shorter than the 60-second monitor tick because a stale holdings set once phantom-closed seventeen real positions. The rolling reject window replaced a session-scoped one that silenced a whole weekend. The per-book position cap replaced a platform-wide count that killed 516 entries in a day.

A number without its reason is a number somebody will optimise later. That is why this document states both.

Prepared by Nova for Mike · commit eb271bb · 23 August 2026. Every threshold in this document was read from the deployed source rather than from design notes, and every claim in Part 7 is drawn from the engine's own audit record.